NSMO of KSA: Stage 1 Screening & Stage 2 Regional Qualifiers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

NSMO of KSA: Stage 1 Screening & Stage 2 Regional Qualifiers Practice Exercises

1 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{15} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 5

b. 4

c. 3 ✓

d. 15

Rationalizing each denominator by multiplying by $\sqrt{k+1}-\sqrt{k}$ gives $\frac{1}{\sqrt{k+1}+\sqrt{k}} = \sqrt{k+1}-\sqrt{k}$. Telescoping from $k=1$ to 15 leaves $\sqrt{16}-\sqrt{1} = 4-1 = 3$.

2 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{24} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 6

b. 4 ✓

c. 5

d. 24

Rationalizing each denominator by multiplying by $\sqrt{k+1}-\sqrt{k}$ gives $\frac{1}{\sqrt{k+1}+\sqrt{k}} = \sqrt{k+1}-\sqrt{k}$. Telescoping from $k=1$ to 24 leaves $\sqrt{25}-\sqrt{1} = 5-1 = 4$.

3 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{35} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 5 ✓

b. 7

c. 6

d. 35

Rationalizing each denominator by multiplying by $\sqrt{k+1}-\sqrt{k}$ gives $\frac{1}{\sqrt{k+1}+\sqrt{k}} = \sqrt{k+1}-\sqrt{k}$. Telescoping from $k=1$ to 35 leaves $\sqrt{36}-\sqrt{1} = 6-1 = 5$.

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4 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{48} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 8

b. 7

c. 48

d. 6 ✓

Rationalizing each denominator by multiplying by $\sqrt{k+1}-\sqrt{k}$ gives $\frac{1}{\sqrt{k+1}+\sqrt{k}} = \sqrt{k+1}-\sqrt{k}$. Telescoping from $k=1$ to 48 leaves $\sqrt{49}-\sqrt{1} = 7-1 = 6$.

5 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{8} \frac{2}{(2k-1)(2k+1)}$.

a. $\frac{16}{15}$

b. $\frac{2}{17}$

c. $\frac{16}{17}$ ✓

d. $\frac{32}{17}$

Using partial fractions, $\frac{2}{(2k-1)(2k+1)} = \frac{2}{2}\left(\frac{1}{2k-1}-\frac{1}{2k+1}\right)$. Telescoping from $k=1$ to 8 gives $\frac{2}{2}\left(1-\frac{1}{17}\right) = \frac{2}{2}\cdot\frac{16}{17} = \frac{16}{17}$.

6 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{9} \frac{3}{(2k-1)(2k+1)}$.

a. $\frac{27}{17}$

b. $\frac{27}{19}$ ✓

c. $\frac{3}{19}$

d. $\frac{54}{19}$

Using partial fractions, $\frac{3}{(2k-1)(2k+1)} = \frac{3}{2}\left(\frac{1}{2k-1}-\frac{1}{2k+1}\right)$. Telescoping from $k=1$ to 9 gives $\frac{3}{2}\left(1-\frac{1}{19}\right) = \frac{3}{2}\cdot\frac{18}{19} = \frac{27}{19}$.

7 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{10} \frac{4}{(2k-1)(2k+1)}$.

a. $\frac{40}{21}$ ✓

b. $\frac{40}{19}$

c. $\frac{4}{21}$

d. $\frac{80}{21}$

Using partial fractions, $\frac{4}{(2k-1)(2k+1)} = \frac{4}{2}\left(\frac{1}{2k-1}-\frac{1}{2k+1}\right)$. Telescoping from $k=1$ to 10 gives $\frac{4}{2}\left(1-\frac{1}{21}\right) = \frac{4}{2}\cdot\frac{20}{21} = \frac{40}{21}$.

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8 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{11} \frac{5}{(2k-1)(2k+1)}$.

a. $\frac{55}{21}$

b. $\frac{5}{23}$

c. $\frac{110}{23}$

d. $\frac{55}{23}$ ✓

Using partial fractions, $\frac{5}{(2k-1)(2k+1)} = \frac{5}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$. Telescoping from $k=1$ to 11 gives $\frac{5}{2}\left(1 - \frac{1}{23}\right) = \frac{5}{2} \cdot \frac{22}{23} = \frac{55}{23}$.

9 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{5} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{30}{31}$

b. $\frac{25}{62}$

c. $\frac{15}{31}$ ✓

d. $\frac{1}{31}$

Since $(k^2 + k + 1) - (k^2 - k + 1) = 2k$, $\frac{k}{k^4 + k^2 + 1} = \frac{1}{2}\left(\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1}\right)$. Notice $(k+1)^2 - (k+1) + 1 = k^2 + k + 1$, so the sum telescopes cleanly from $k=1$ to 5: $\frac{1}{2}\left(1 - \frac{1}{5^2 + 5 + 1}\right) = \frac{15}{31}$.

10 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{6} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{42}{43}$

b. $\frac{21}{43}$ ✓

c. $\frac{18}{43}$

d. $\frac{1}{43}$

Since $(k^2 + k + 1) - (k^2 - k + 1) = 2k$, $\frac{k}{k^4 + k^2 + 1} = \frac{1}{2}\left(\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1}\right)$. Notice $(k+1)^2 - (k+1) + 1 = k^2 + k + 1$, so the sum telescopes cleanly from $k=1$ to 6: $\frac{1}{2}\left(1 - \frac{1}{6^2 + 6 + 1}\right) = \frac{21}{43}$.

11 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{7} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{28}{57}$ ✓

b. $\frac{56}{57}$

c. $\frac{49}{114}$

d. $\frac{1}{57}$

Since $(k^2 + k + 1) - (k^2 - k + 1) = 2k$, $\frac{k}{k^4 + k^2 + 1} = \frac{1}{2}\left(\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1}\right)$. Notice $(k+1)^2 - (k+1) + 1 = k^2 + k + 1$, so the sum telescopes cleanly from $k=1$ to 7: $\frac{1}{2}\left(1 - \frac{1}{7^2 + 7 + 1}\right) = \frac{28}{57}$.

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12 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{8} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{72}{73}$

b. $\frac{32}{73}$

c. $\frac{1}{73}$

d. $\frac{36}{73}$ ✓

Since $(k^2 + k + 1) - (k^2 - k + 1) = 2k$, $\frac{k}{k^4 + k^2 + 1} = \frac{1}{2}\left(\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1}\right)$. Notice $(k+1)^2 - (k+1) + 1 = k^2 + k + 1$, so the sum telescopes cleanly from $k = 1$ to 8: $\frac{1}{2}\left(1 - \frac{1}{8^2 + 8 + 1}\right) = \frac{36}{73}$.

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